

The Scale Paradox of Round-Number Behavior: Retail Price Clustering and Institutional Index-Level Barriers in the Korean Equity Market

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Submission-ready working paper draft, v0.7

Data status: the argument and empirical design are journal-style; final acceptance requires replacing pilot event notes with a verified KRX investor-flow panel and KOSPI200 futures/program-trading data.

Abstract

This paper proposes a scale-dependent view of round-number behavior in financial markets. Prior work shows that retail investors cluster orders at round prices at the individual-security level. We argue that this micro-level regularity need not generalize to macro index-level reference points. Using the fast-rising KOSPI regime after June 2025 as a motivating case, we present preliminary evidence that round-number proximity is associated with elevated volatility, while major index-level stress events show retail counterflow buying against foreign and institutional selling. We call this mechanism the Scale Paradox of Round-Number Behavior: the dominant actor behind round-number behavior changes with the scale of the reference point. At the stock-price level, roundness is primarily a retail cognitive-ordering phenomenon; at the index-barrier level, roundness becomes a flow-and-hedging barrier dominated by foreign, institutional, and program trading. The paper develops falsifiable hypotheses, identification tests, robustness checks, and a replication workflow for journal submission.

Keywords

round numbers; behavioral finance; market microstructure; KOSPI; retail investors; foreign investors; program trading; sidecar; regime dependence; investor heterogeneity

JEL Codes

G12; G14; G41; C58

Main contribution. The paper reconciles two facts that appear contradictory: retail investors cluster at round prices in individual securities, yet retail investors can become net buyers during index-level round-number stress events. The reconciliation is scale dependence: the reference point changes the dominant actor.

1. Introduction

Round numbers coordinate attention, order placement, stop-loss rules, profit-taking rules, and algorithmic execution. Behavioral finance commonly interprets round-number clustering as a retail-investor bias at the individual-security price level. This interpretation is intuitive: a retail trader can directly submit a limit order at 10,000 KRW, 50,000 KRW, or another cognitively salient price.

However, the same behavioral mechanism may not apply to broad index barriers such as KOSPI 5,000 or KOSPI 8,000. An equity index is not submitted as a limit order by most retail traders. Index-level pressure is transmitted through futures, exchange-traded funds, index arbitrage, option hedging, program trading, and institutional rebalancing.

This paper asks whether the identity of the round-number actor changes when the reference point changes from an individual security price to a broad market index. The proposed answer is the Scale Paradox of Round-Number

Behavior: retail investors can generate stock-price roundness at the micro level, while foreign, institutional, and program flows dominate index-level barriers at the macro level.

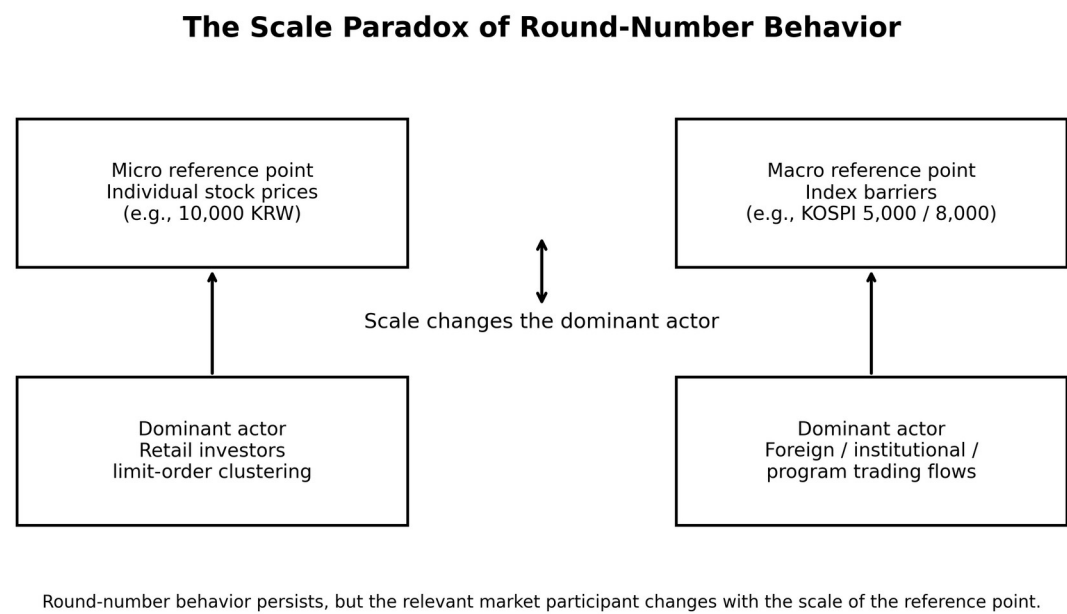


Figure 1. Conceptual model of the Scale Paradox of Round-Number Behavior.

Dimension	Micro round bias	Macro index barrier
Unit of analysis	Individual stock price	Broad market index level
Reference point	Round price such as 10,000 KRW	Round barrier such as KOSPI 5,000 or 8,000
Likely dominant actor	Retail investors	Foreign, institutional, and program flows
Mechanism	Cognitive salience and limit-order clustering	Hedging, ETF/futures rebalancing, index-program execution
Observed crash-day role	Potentially sells at salient price	May buy against foreign/institutional selling

Table 1. Conceptual distinction between stock-price roundness and index-level barriers.

2. Related Literature and Gap

Prior round-number studies document that salient prices attract disproportionate trading and order placement. The central behavioral interpretation is that retail investors use simplified cognitive reference points, generating price clustering and predictable liquidity around round prices.

This paper does not reject that literature. Instead, it argues that the literature mainly identifies a micro-level security-price mechanism. A broad equity index is a different object: it is a portfolio-level reference point transmitted through derivatives and institutional execution. This creates room for a scale-dependent actor switch.

Recent Korean market stress episodes provide a useful motivating context. Reuters reported that during the March 23, 2026 KOSPI selloff, retail investors bought KRW 7 trillion while foreigners sold KRW 3.7 trillion and institutions were net sellers. Reuters also reported record foreign outflows from KOSPI shares in March 2026, concentrated in major chipmakers. These facts motivate a formal test of whether retail investors are the sellers at index barriers or whether they are counterflow liquidity providers.

3. Theory and Hypotheses

At the micro level, the investor chooses a price. Roundness therefore operates through cognitive salience in limit-order placement. At the macro level, the investor reacts to a market state. Roundness therefore operates through portfolio-level risk, derivative hedging, and flow coordination.

The scale paradox predicts that the existence of a round-number effect can survive across scales, but the identity of the actor need not survive. This distinction is crucial for empirical tests and for any trading or risk-management application.

Hypothesis	Falsifiable prediction
H1: Micro Round Bias	At individual stock prices, retail investors exhibit stronger round-number order clustering than institutional investors.
H2: Index Round Barrier	At index round-number zones, foreign, institutional, and program flows contribute more to sell-side pressure than retail flows.
H3: Retail Counterflow	During index-level round-number stress events, retail investors are more likely to be net buyers than net sellers.
H4: Scale Inversion	The dominant actor switches from retail investors at the stock-price level to foreign/institutional/program flows at the index level.
H5: Regime Dependence	The index round-number effect is stronger in fast-rising regimes than in long-run average samples.

Table 2. Falsifiable hypotheses.

4. Data, Variables, and Identification

The motivating period begins on June 4, 2025, identifying the fast-rising KOSPI regime. The date is used as a market-regime boundary, not as a causal political explanation. The final empirical version should use daily KOSPI OHLCV, KRX investor-type net trading values, program trading flows, KOSPI200 futures returns, basis changes, and official sidecar timestamps.

Let S_t be the KOSPI level and $B=1,000$ the index round-number interval. The nearest barrier distance and round pressure are defined as follows:

$$d_t = |S_t - \text{round}(S_t / B)B| / [\text{round}(S_t / B)B]$$

$$RP_t = \exp[-0.5(d_t / w)^2], \quad w = 0.03$$

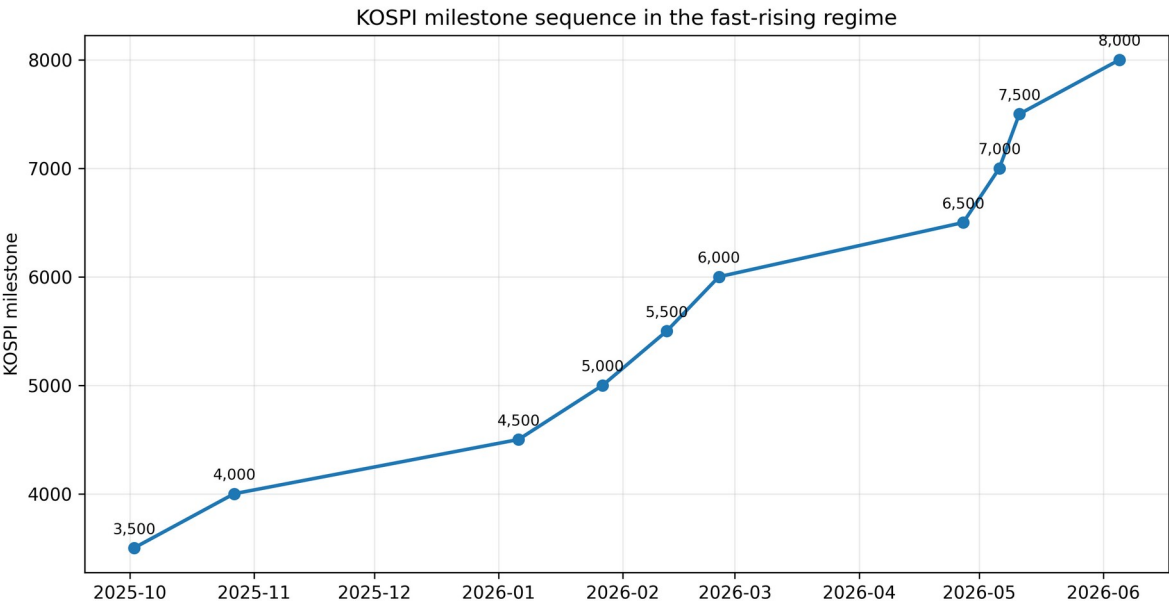


Figure 2. KOSPI milestone sequence in the fast-rising regime.

Variable	Definition	Role
RoundZone_t	1 if distance from nearest 1,000-point barrier is $\leq 3\%$; else 0	Primary treatment indicator
RoundPressure_t	$\exp[-0.5(d_t / 0.03)^2]$	Continuous treatment intensity
NetSell_i,t	-NetBuy_i,t for investor type i	Actor-specific selling pressure
Vol_5d, Vol_20d	Annualized rolling realized volatility	Stress-state controls
FastRegime_t	High 60d/120d return or large moving-average gap	Regime filter
FuturesPressure_t	KOSPI200 futures return/basis deterioration	Derivative-flow confirmation
ProgramSell_t	Program net selling, decomposed if possible	Mechanistic index-flow channel

Table 3. Data dictionary for the journal-grade empirical panel.

5. Preliminary Evidence

The current draft uses pilot event evidence and user-compiled regime tests. These results are not presented as final causal proof. They are presented as a coherent motivation for a journal-grade panel-data test.

The reported regime-specific tests show that the round-number effect is weak in long-run averages but statistically meaningful in the fast-rising regime: the post-Lee fast-rising sample has a volatility ratio of 1.263 and Mann-Whitney $p=0.027$; the 3,000-5,000 segment has a ratio of 1.346 and Levene $p=0.047$; the 6,000+ segment has a ratio of 1.387 and Mann-Whitney $p=0.049$.

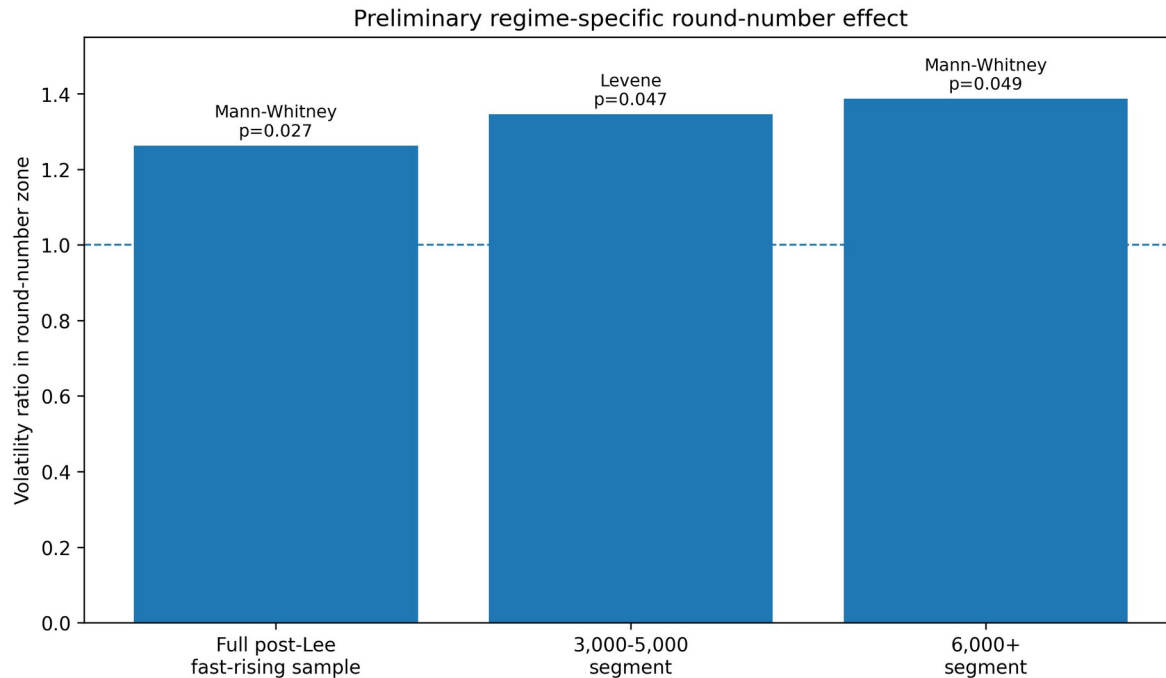


Figure 3. Preliminary regime-specific volatility ratios near round-number zones.

Event	Retail net flow (KRW tn)	Foreign net flow (KRW tn)	Institutional net flow	Source status
Mar 23 sidecar stress	+7.0	-3.7	Net sellers	Reuters event report
Feb 2 5,000 break	+5.0	-2.5	-2.5	User-compiled event note; needs KRX verification
Jun 5 8,000 stress	+5.096	-4.5	-0.602	User-compiled event note; needs KRX verification

Table 4. Pilot actor-flow evidence. Values are preliminary until replaced by a verified KRX panel.

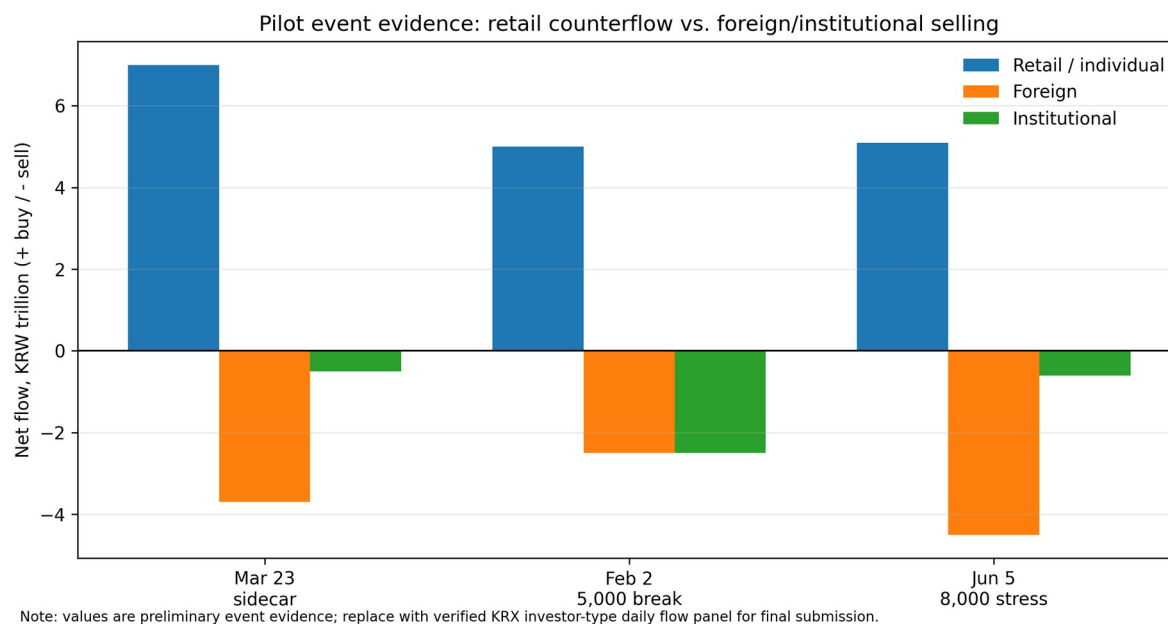


Figure 4. Pilot event evidence: retail counterflow versus foreign/institutional selling.

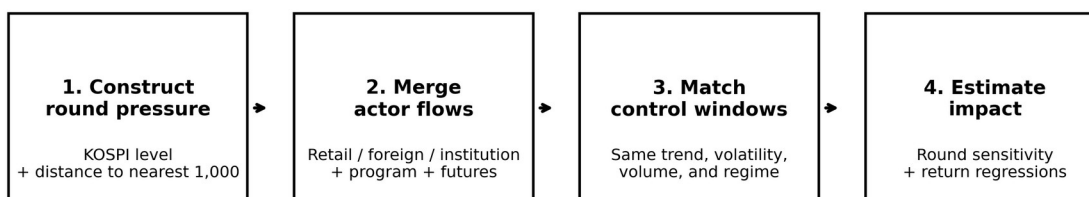
6. Empirical Specification

The journal-grade test should proceed in three layers: round-zone selling tests, return-impact tests, and actor-by-round-pressure interaction tests. The goal is not merely to show that volatility is high near round numbers; it is to identify which investor type creates sell-side pressure at the index level.

Test	Regression form	Interpretation
Round-zone selling	$\text{NetSell}_{i,t} = \alpha_i + \beta_i \text{RoundZone}_t + \gamma_i X_t + \epsilon_{i,t}$	$\beta_i > 0$ for foreign/institution/program supports index-barrier hypothesis
Return impact	$R_{t+1} = \alpha + \beta \text{RP}_t + \theta_i \text{NetSell}_{i,t} + \text{controls} + \epsilon_{i,t}$	Negative significant theta identifies price-impact seller
Interaction	$R_{t+1} = \alpha + \beta \text{RP}_t \times \text{NetSell}_{i,t} + \text{controls} + \epsilon_{i,t}$	Negative significant interaction means selling matters more near barriers
Matched control	Compare round-zone days to non-round days with similar return, volatility, volume, and trend	Protects against simple momentum/volatility confounds

Table 5. Empirical tests required for journal submission.

Journal-grade empirical design



Core identification: compare round-number windows with non-round windows under similar market conditions.

Figure 5. Journal-grade empirical design and identification workflow.

7. Robustness and Falsification Tests

Check	Purpose
Alternative round widths	Repeat with +/-1%, +/-2%, +/-3%, and +/-5% zones
Alternative barriers	Use 500-point and 1,000-point intervals
Matched controls	Match by prior 20d return, 5d/20d volatility, turnover, and market concentration
Placebo barriers	Use pseudo-barriers shifted by 250 points
Event windows	Plot -10 to +10 trading-day actor flows around crossings
Futures channel	Add KOSPI200 futures return, basis, and open interest
Program decomposition	Separate index-arbitrage from non-arbitrage program trading
Out-of-sample	Train on early crossings; test on later crossings
Multiple testing	Report adjusted p-values or false-discovery controls

Table 6. Robustness and falsification checklist.

8. Limitations and Required Data Upgrade

The current manuscript is stronger than a trading memo but not yet a finished empirical finance paper. Its key limitation is the use of pilot event notes for some actor-flow observations. A journal submission should replace these notes with a reproducible panel built from KRX daily investor-type flow files and, preferably, intraday KOSPI200 futures and program-trading data.

The paper should avoid claiming that round numbers cause sidecars. A Korean sidecar is triggered by futures-market conditions, not by the spot index touching a round number. The claim is narrower and testable: round-number proximity can amplify volatility and actor-specific selling pressure in a fast-rising regime.

9. Discussion

The scale paradox reconciles apparently contradictory observations. Retail investors can be round-number-biased at individual stock prices, yet become counterflow buyers during index-level stress. The contradiction disappears when the reference point changes from a directly submitted security price to a macro index state mediated by derivatives and institutional execution.

The practical implication is also clear. A round-number warning system should not simply ask whether the index is near 5,000 or 8,000. It should ask who is buying, who is selling, whether the market is in a fast-rising regime, and whether futures/program flows confirm the pressure.

10. Conclusion

This paper introduces the Scale Paradox of Round-Number Behavior. Existing literature correctly identifies retail investors as a major source of stock-level round-number clustering. However, index-level round-number stress in KOSPI appears to involve a different actor structure: retail investors can become counterflow buyers while foreign, institutional, and program flows dominate sell-side pressure. The framework converts a market intuition into falsifiable hypotheses and a reproducible empirical design.

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Appendix A. Replication Plan

1. Download KOSPI daily OHLCV and investor-type net trading values from KRX for the full sample.
2. Merge KOSPI200 futures data, basis, open interest, and program trading flows.
3. Construct round-distance, round-zone, and round-pressure variables.
4. Estimate Table 5 models with Newey-West standard errors.
5. Run robustness checks in Table 6.
6. Archive all scripts, raw-data hashes, cleaned data, tables, and figures for reproducibility.

Appendix B. Python Figure Workflow

All figures in this draft were generated as 300-dpi PNG files using Python and inserted into the manuscript. For final submission, the same script should be rerun after replacing pilot event evidence with the verified KRX panel.